

Individual Differences in Visual Masking Depend Only on the Stimulus Type, Not on
Masking Type

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Abstract

ABSTRACT

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Introduction

Understanding how fundamental perceptual abilities interrelate and contribute to cognitive performance is one of the central topics in cognitive psychology and individual-differences research. Perhaps the best-known finding in this literature concerns inspection time (IT), the minimum exposure duration at which an observer can reliably make a simple perceptual discrimination. “Perceptual Indices of Performance” (n.d.) first introduced this paradigm. In its classic form, participants briefly view a stimulus consisting of two vertical lines of clearly unequal length joined at the top (hereafter, the Pi stimulus) and indicate which line is longer. By varying the exposure duration of the Pi stimulus and using a backward mask to prevent further visual processing, the minimum exposure duration required for accurate discrimination can be estimated. Nettelbeck and Lally (1976) subsequently reported that IT correlated negatively with Performance IQ. This negative relation has since been replicated many times across age groups, sensory modalities, and measures of intelligence (Deary, Caryl, Egan, & Wight, 1989; Deary & Stough, 1996; Osmon & Jackson, 2002). Meta-analyses place the uncorrected correlation between IT and intelligence at about $-.3$, rising to about $-.5$ after correction for measurement error and range restriction (Grudnik & Kranzler, 2001; Kranzler & Jensen, 1989). The standard interpretation of this association is in terms of *processing speed*. Because the mask interrupts further processing of the target, performance depends on how much target information the observer can extract before the mask arrives; individuals who process information faster therefore require shorter exposures to discriminate the stimulus accurately (Vickers & Smith, 1986). Faster information processing, in turn, may benefit cognitive performance across a wide range of tasks (Vernon, 1983), resulting in negative correlations between IT and cognitive performance.

Backward masking is therefore a key manipulation for limiting how much target information can be extracted. As the interval between target onset and mask onset shortens, only observers who process information quickly can extract enough information from the target before the mask disrupts further processing. The situation changes, however, when target and mask are presented at the same time and remain on the screen for a sufficient duration. The target is then embedded in the mask rather than interrupted by it, a procedure known as *simultaneous masking*. In simultaneous masking, processing time is no longer the limiting factor; performance instead depends on how precisely the relevant features of the target can be extracted from the concurrent noise. This taps the same sensory discrimination ability assessed by classic sensory discrimination tasks, in which observers judge which of two stimuli is higher in pitch or longer in length (Deary, 1994; Deary, Bell, Bell, Campbell, & Fazal, 2004). In both cases, performance is limited by how precisely the signal is represented relative to noise, whether that noise is added to the display or arises within the sensory system (Pelli & Farell, 1999). We therefore treat simultaneous masking as a measure of sensory discrimination and use this term throughout the remainder of the paper.

Like processing speed, sensory discrimination has long been linked to cognitive performance. Galton (1883) proposed that finer sensory acuity supplies the raw material for intelligent judgment, and Spearman (1904) found that discrimination of pitch, light, and weight loaded on a general factor of intelligence, which he termed *g*. Subsequent studies have reported reliable correlations between single discrimination tasks and intelligence. (Mikellidou, Lambrou, Georgiou, & Avraamides, 2023; Mosing, Pedersen, Madison, & Ullén, 2014; Raz, Willerman, & Yama, 1987). Deary et al. (2004) reported a latent correlation of .92 between general discrimination and general intelligence, and Tsukahara, Harrison, Draheim, Martin, and Engle (2020) found latent correlations of .71 between sensory discrimination and attention control.

Although both processing speed and sensory discrimination have shown reliable

correlations with cognitive performance, few studies have measured them in the same participants, and none, to our knowledge, has done so with the same stimuli and masks. The closest precedent is Tsukahara et al. (2020), which measured both constructs in the same participants. However, their sensory discrimination tasks required auditory judgments of the pitch, loudness, or duration of tones, whereas their processing speed tasks required speeded comparisons of visually presented letter and digit strings. When stimulus, sensory modality, and construct vary together, two alternative interpretations cannot be ruled out. First, individual differences may be partly stimulus-specific. Studies that administer batteries of visual tasks to the same observers typically find weak correlations among tasks (Cappe, Clarke, Mohr, & Herzog, 2014; Tulver, 2019), so an observer may perform well with one type of stimulus but poorly with another. An apparent dissociation between processing speed and sensory discrimination may then reflect differences between the materials rather than between the processes. Second, the variance that the two kinds of tasks share may reflect a general perceptual factor rather than two distinct abilities (Deary, 1994), in which case the distinction between sensory discrimination and processing speed would not hold at the level of individual differences. In the present study, we addressed both possibilities by holding the stimulus constant while varying only the masking procedure. We used three stimulus types, a Gabor patch, a letter, and the Pi stimulus, and each was presented with an identical mask under both backward masking and simultaneous masking. This fully crossed 2 (masking procedure) $\times$ 3 (stimulus type) design follows the logic of the multitrait-multimethod matrix (Campbell & Fiske, 1959): because every stimulus appears under both masking procedures, variance associated with the masking procedure can be separated from variance associated with the stimulus. If sensory discrimination and processing speed are distinct abilities, tasks that share a masking procedure should correlate more strongly than tasks that share a stimulus; if individual differences are instead organized by stimulus content, the reverse should hold.

Participants also completed three complex span tasks of working memory (Conway et

al., 2005; Pahor, Seitz, & Jaeggi, 2022; Unsworth, Heitz, Schrock, & Engle, 2005) and a matrix reasoning test of fluid intelligence (Pahor, Stavropoulos, Jaeggi, & Seitz, 2019), which together served as our measures of cognitive performance. This allowed us to ask, in addition, whether sensory discrimination and processing speed share their relation to cognitive performance, and whether that relation depends on stimulus content.

Task description

To assess the general discrimination ability and processing speed, the experiment contained six visual masking tasks: three employing spatial masking 1 and three employing temporal masking 2. Both sets of tasks used the same stimulus classes (a Gabor stimulus, a letter stimulus, and a Pi stimulus) along with the masks and task requirements. The only difference was the masking procedure: in the spatial masking tasks, the stimulus and mask were presented simultaneously, whereas in the temporal masking tasks the mask followed the stimulus after a variable stimulus-onset asynchrony (SOA). This design ensured stricter control of experimental variables, preventing any potential confounds other than the spatial-temporal manipulation from influencing the results. Each task began with 25 practice trials at higher visibility, during which accuracy feedback was provided. The main experimental phase then followed without feedback. For all six tasks, stimulus contrast was adaptively controlled using a two-down-one-up (2D1U) staircase procedure, which adjusted task difficulty based on participants' performance and converged to approximately 70.7% accuracy. Each task comprised eight blocks of 25 trials (200 trials in total). From the second block onward, the initial contrast level was set to the final contrast level obtained at the end of the preceding block, allowing task difficulty to adapt across blocks.

Need description for cognitive tasks

Sensory discrimination tasks

Gabor. Participants completed a two-alternative forced-choice orientation discrimination task. On each trial, a fixation cross was presented for 485 ms, followed by a centrally presented, random-phase Gabor patch ($\pm 45^\circ$ orientation) embedded in white noise, which remained on the screen for another 485 ms. Participants indicated whether the grating was tilted up-left or up-right by pressing the keyboard. Task difficulty was adaptively adjusted by varying stimulus contrast.

Letter. Participants performed a forced-choice letter identification task using the keys A–L (second keyboard row). Each trial began with a 485 ms fixation cross, followed by a centrally presented target letter (one of A, S, D, F, G, H, J, K, L) displayed for 485 ms. The letter was embedded in a frequency-matched letter noise mask constructed as follows: We constructed the mask by first computing the pixelwise average of the nine possible letters and then perturbing its Fourier spectrum with complex Gaussian noise. After inverse transformation and normalization, this yielded a structured noise pattern with the same overall spatial frequency profile as the average letter image. On each trial, the target letter was linearly blended with this mask (blend coefficient α), and participants identified the letter using the corresponding keyboard key. Task difficulty was adaptively adjusted by varying α .

Pi. Participants performed a forced-choice judgment on a Π -shaped (Pi) target, indicating which vertical “leg” was longer (left vs. right). Each trial began with a 485 ms fixation cross, followed by a centrally presented Π stimulus for 485 ms. The mask was composed of random oblique line segments arranged to form a Π -like clutter. Short line elements (randomized in length and orientation within $\pm 10^\circ$ – 45°) were placed in two vertical columns aligned with the legs of the Π and around the crossbar region. Equal numbers of lines were drawn on each side, although their random placement could introduce local asymmetries. On each trial, the target Π was superimposed on this

cluttered mask, and participants indicated which leg appeared longer using the keyboard. Task difficulty was adaptively controlled by varying the leg-length difference (ΔL).

Processing Speed Tasks

Gabor. The task matched the spatial Gabor-orientation discrimination task. Each trial began with a 485 ms fixation cross, followed by a centrally presented, random-phase Gabor patch ($\pm 45^\circ$ orientation) embedded in white noise. After a variable SOA, the target was replaced by a dynamic sequence of five white noise masks (48 ms each). Participants indicated whether the grating was tilted up-left or up-right by pressing the keyboard. Task difficulty was adaptively adjusted by SOA.

Letter. The task matched the spatial letter-identification condition. Each trial began with a 485 ms fixation cross, followed by a centrally presented target letter (one of A, S, D, F, G, H, J, K, L). After a variable SOA, the target was replaced by a dynamic sequence of five frequency-matched letter noise masks (48 ms each). Participants identified the letter using the corresponding keyboard key. Task difficulty was adaptively adjusted by varying SOA.

Pi. The task matched the spatial II-leg discrimination condition. Each trial began with a 485 ms fixation cross, followed by a centrally presented II stimulus. After a variable SOA, the target was replaced by a dynamic sequence of five clutter masks composed of randomly oriented oblique line segments (48 ms each). Participants indicated which leg appeared longer using the keyboard. Task difficulty was adaptively controlled by varying SOA.

Cognitive tasks

OSPAN. Participants completed an automated OSPAN task (Unsworth et al., 2005) combining simple math operations with working memory recall for letter sequence.

Each set contained 3–7 letters, with each set size presented three times, yielding 15 memory sets in total. Before the main task, participants completed three practice sets of size three to familiarize themselves with the procedure. In addition, they solved 10 math problems beforehand to establish individual processing speed. In the main task, set sizes 3–7 were each presented three times in randomized order (15 sets total). Within each set, trials alternated between (i) a math operation screen (e.g., “ $6 + 2 = ?$ ”), after which the participant clicked to continue, and (ii) a proposed answer screen showing a single digit; participants judged whether the digit correctly solved the operation by clicking true/false. A single uppercase letter was then shown for 800 ms, and the sequence continued until the set ended. At recall, participants clicked the letters in serial order from a matrix. For each set, participants earned points equal to the set size only if (a) all math operations are solved less than their average solving time in practice trials plus 2.5 SD, (b) no math errors occurred for that set, and (c) all letters were recalled in the correct order. Sets failing either criterion scored 0. The final OSPAN score was the sum of points across all sets.

CSPAN. Participants completed a CSPAN task (Conway et al., 2005) combining serial counting with working memory recall for number sequence. Each display contained a set of target items (green circles) intermixed with distractors (yellow circles). Participants counted the number of targets in each display, after which the participant clicked to continue to the next display. At recall, participants were prompted to recall the sequence of counts (e.g., “3–5–2”) in the exact order they were encountered. Responses were made by clicking the digits from a matrix. The procedure began with three practice sets of size 2. The main task then presented set sizes 3–7, each three times in randomized order (15 sets total). A set contributed points equal to its size only if the final recall sequence was accurate in both values and order. The final CSPAN score was the sum of points across all sets.

Complex Corsi Participants completed a SSPAN task adapted from the Corsi block-tapping paradigm to assess both storage and processing aspects of spatial working

memory (Pahor et al., 2022). On each trial, cartoon gophers appeared sequentially in one of 12 possible screen locations, each presented for 1,500 ms with a 500 ms inter-stimulus interval. Participants were required to reproduce the sequence by tapping the corresponding locations in the exact order. Between each gopher presentation participants performed a secondary sorting task, dragging an item to the left or right, thereby imposing concurrent processing demands and limiting reliance on mnemonic strategies. Sequence length ranged adaptively from 2 to 10 items, increasing or decreasing in difficulty depending on accuracy. The final SSPAN score was based on a partial-credit unit method, where each correctly recalled location in the correct serial position contributed one point.

UCMRT. Participants completed the UCMRT (Pahor et al., 2019), a measure of fluid intelligence modeled after Raven’s Progressive Matrices. Each trial presented a 3×3 matrix of abstract visual patterns with the bottom-right cell missing, and participants selected the option that best completed the matrix from eight alternatives displayed below. UCMRT was chosen because it was especially developed for young adults with above-average performance on matrix reasoning tasks, providing a more sensitive and efficient assessment of individual differences in fluid intelligence in this population. The task consisted of 23 problems of varying difficulty, including 2 two-relation items, 3 three-relation items with a single transformation, 6 three-relation items with two transformations, 6 three-relation items with three transformations, and 6 logic items. Performance was scored as accuracy, defined by the number of correctly solved matrices. In addition, the UCMRT is brief, requiring only about 15 minutes to administer, thereby minimizing fatigue and reducing interference with other tasks.

Random-effect psychophysics model

Individual thresholds on the perceptual tasks and the correlations among them were estimated with a hierarchical psychometric model (Rouder & Mehrvarz, 2026). The model estimates individual thresholds and their correlations simultaneously from trial-level data,

thereby accounting for trial-to-trial noise and avoiding the attenuation of correlations caused by measurement error.

At data level, let y_{ijk} denote the binary correctness outcome for the k -th trial of participant i in task j . We assume

$$y_{ijk} \sim \text{Bernoulli}(F(d_{ijk}; \theta_{ij}, \beta_j)),$$

where the success probability is given by a task-specific psychometric function modeled as a log normal CDF.

$$F(d_{ijk}; \theta_{ij}, \beta_j) = \Phi\left(\log\left(\left(\frac{d_{ijk}}{\theta_{ij}}\right)^{\beta_j}\right)\right).$$

Here, $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. The variable d_{ijk} represents the stimulus level: SOA in temporal masking tasks or stimulus strength in spatial masking tasks. The parameter θ_{ij} is the latent threshold of participant i on task j , and β_j controls the slope of the psychometric function for task j . In practice, β_j might need to be fixed in advance to avoid a trade-off between β_j and θ_{ij} depending on the data size and quality (Rouder & Mehrvarz, 2026).

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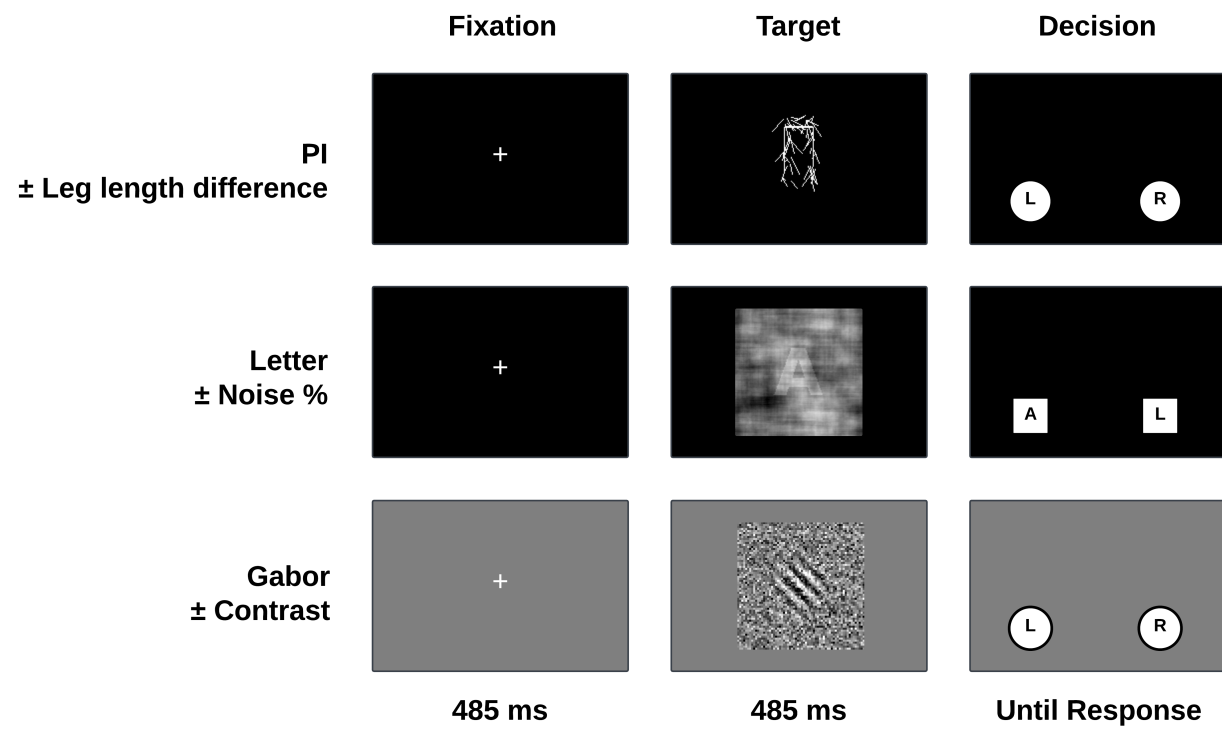


Figure 1. Spatial visual masking tasks.

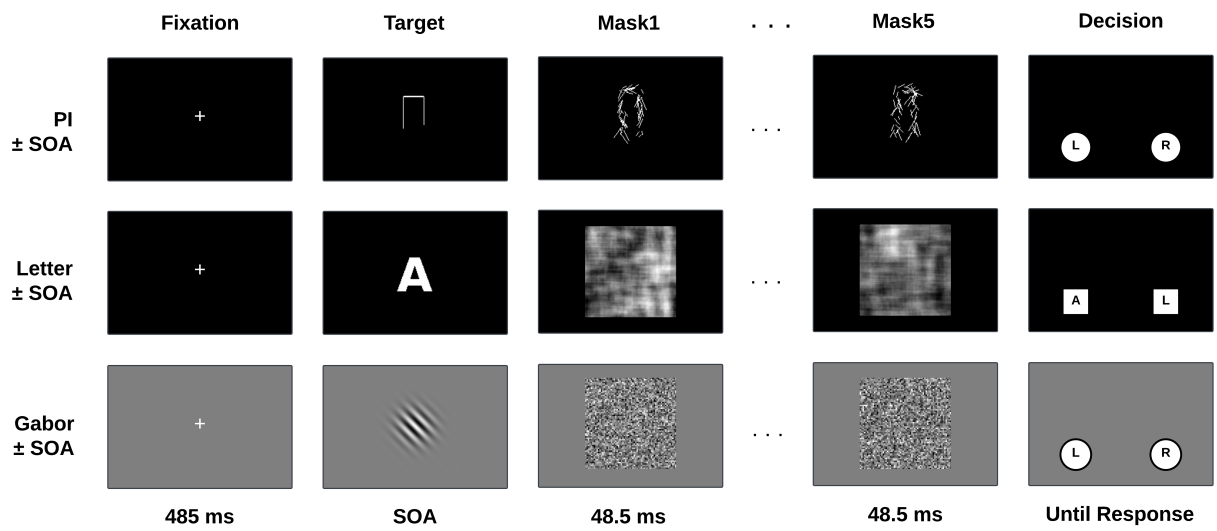


Figure 2. Temporal visual masking tasks.